RESEARCH · 01

Should the Fouled Player Take the Penalty *Himself*?

A ONE-TAILED Z-TEST · TOP 5 EUROPEAN LEAGUES

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renball

TL;DR

Three *sentences*

THE QUESTION

When a goalkeeper fouls a player who then takes the penalty himself, does conversion drop?

THE TEST

One-tailed two-proportion z-test. Treatment n = 79 (75.95%). Control n = 3,456 (78.96%).

THE RESULT

Δ = −3.01 pp (95% CI −13.6 to +5.2) — predicted direction, but p = 0.258: not detectable. Every check agrees (−3.25 pp after controls).

Why we'd *expect* a drop

The just-fouled player walks straight to the spot.

Theory of the taker: physically jarred — body contact, hit the ground, adrenaline. Emotionally agitated, narrowed focus.

Theory of the keeper: locked in. The foul that gave the penalty was his fault. Nothing left to lose. Maximum motivation.

→ *If both pull conversion down, the bench should hand the ball to a designated specialist.*

DATA

What we built the test on

4,110	3,535	79	5
PENALTIES (LOOSE FILTER)	LINKED PENALTIES	TREATMENT-CELL N	LEAGUES

Seasons in scope: 2017–2018 → 2023–2024

Competitions: Premier League, La Liga, Serie A, Bundesliga, Ligue 1.

Source: FBref player-match data, consolidated locally as FULL_DICT_ROWS.pkl.

Linked = the team took exactly one penalty in the match, so taker, fouled player and conceder are unambiguous.

METHOD

One-tailed two-proportion z-test

TREATMENT

GK fouled the taker

Penalties where the goalkeeper conceded the foul AND the fouled player took the penalty himself.

CONTROL

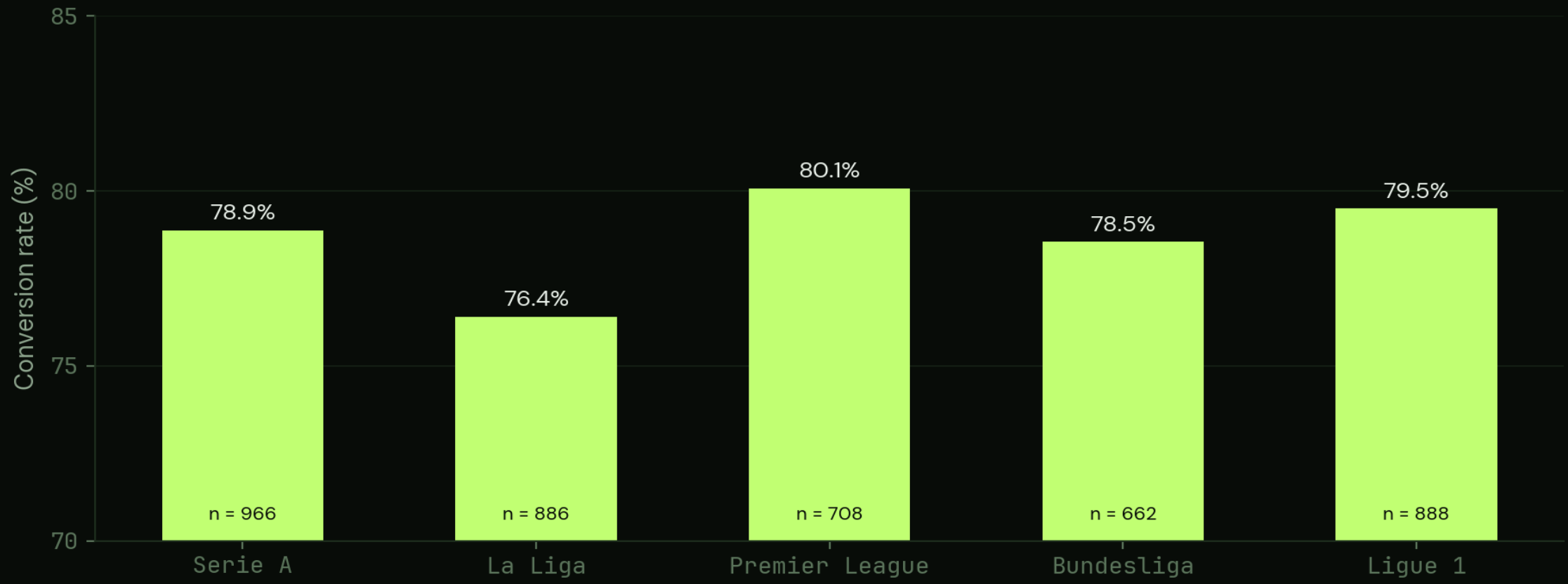
All other penalties

Every other linked penalty — GK conceded but someone else took it, outfielder conceded (any taker), or no foul recorded (e.g. handball).

H_0: conversion_treatment = conversion_control · H_1: conversion_treatment < conversion_control (one-tailed, α = 0.05) · 95% Wilson CIs · statsmodels.proportions_ztest

CONTEXT

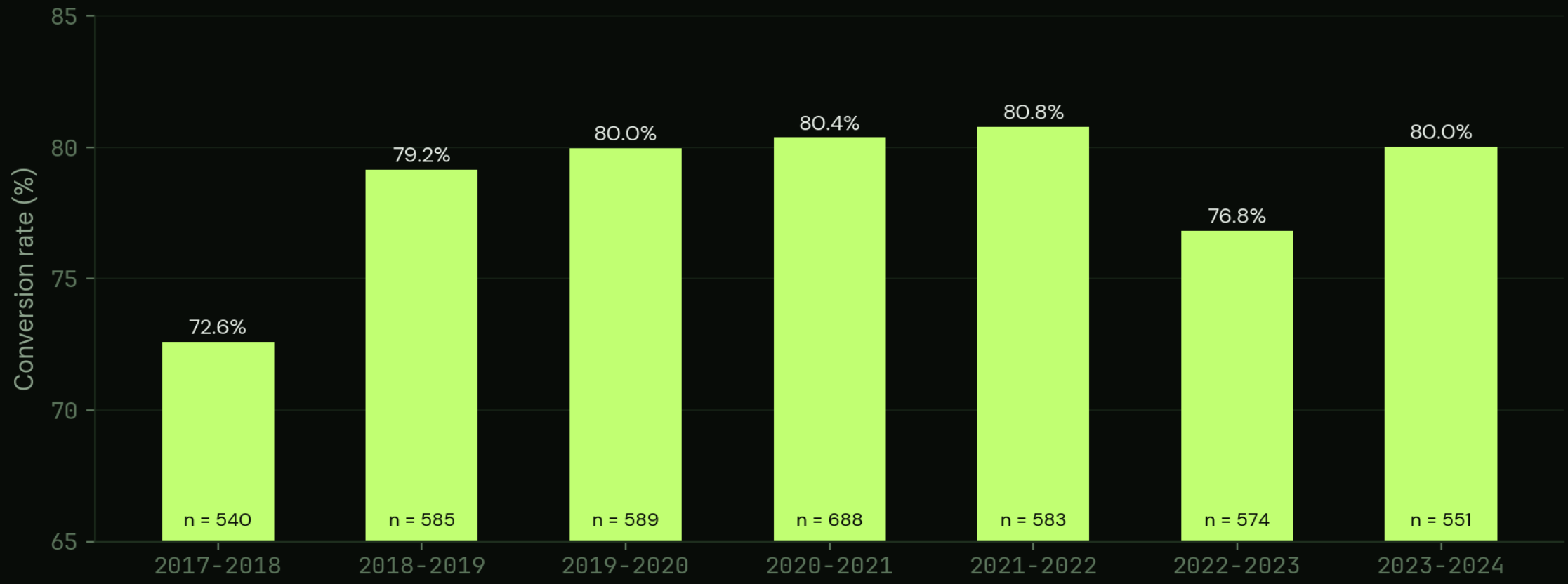
Conversion rate by league



Across the top 5 leagues conversion clusters tightly in the high 70s — league of origin is not a major source of variation.

CONTEXT

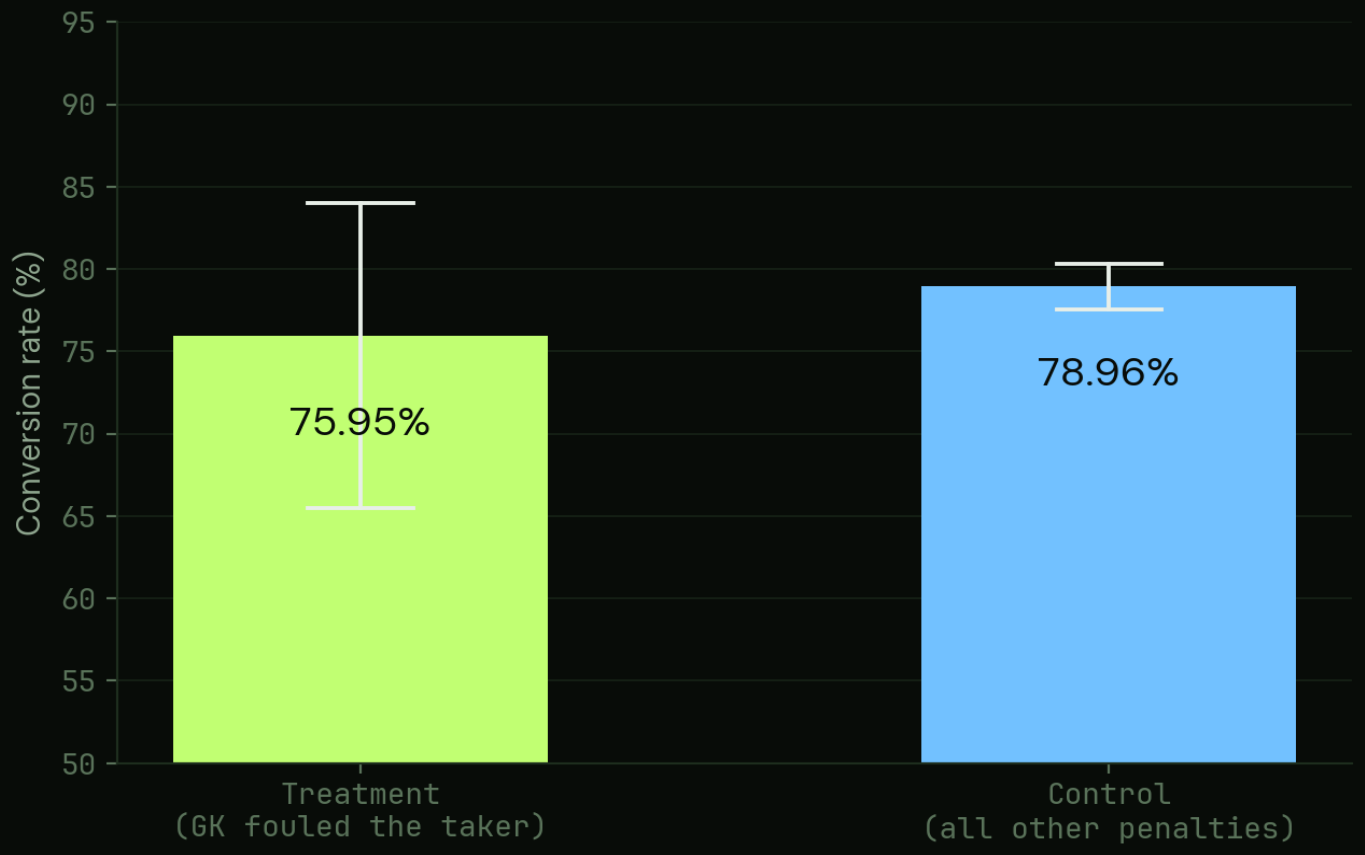
Conversion across seasons



Season rates run from 72.59% (2017-2018) to 80.79% (2021-2022); the lowest is dropped in a sensitivity check.

PRIMARY TEST · ONE-TAILED

Treatment vs *control*



TREATMENT

75.95%

n = 79 · 95% CI [65.46%, 84.03%]

CONTROL

78.96%

n = 3,456 · 95% CI [77.57%, 80.29%]

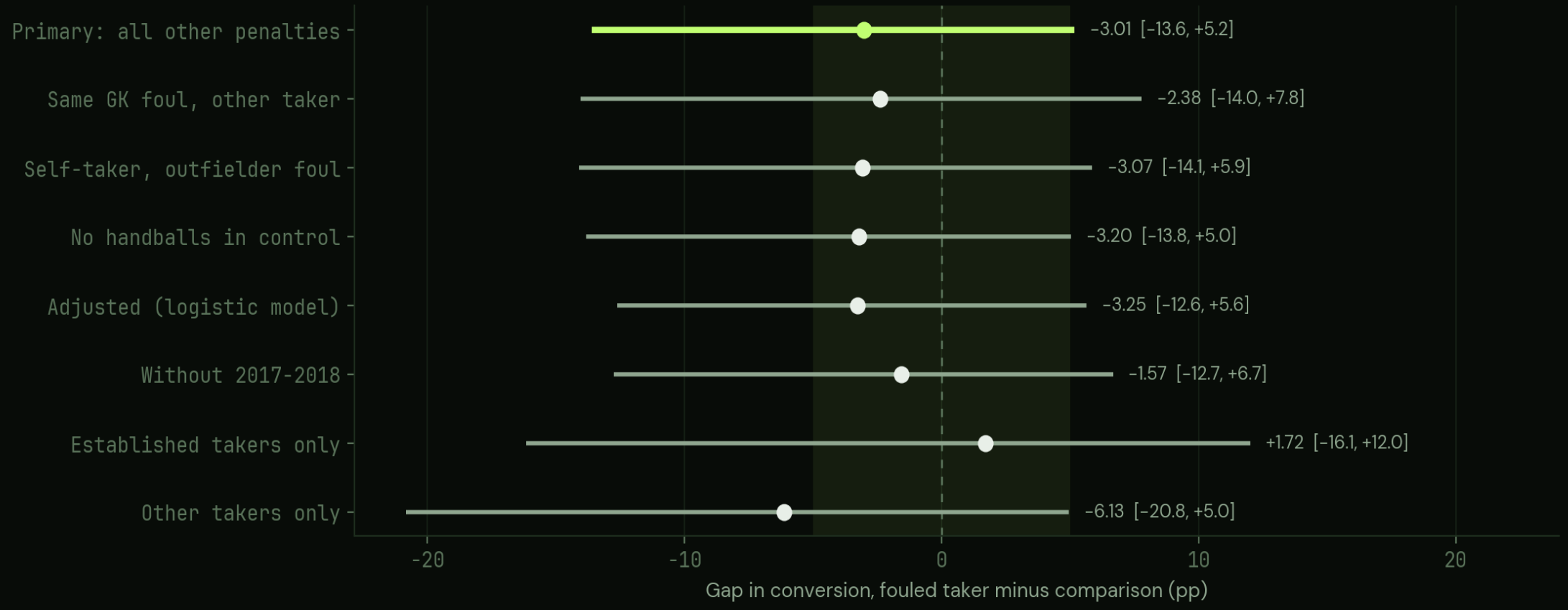
Δ = -3.01 pp z = -0.649

p = 0.2581 (one-tailed) · Fail to reject H_0

Whiskers: 95% Wilson CIs. Difference 95% CI -13.6 to +5.2 pp; smallest detectable drop ≈ 12.6 pp (80% power).

ROBUSTNESS

Does it *hold up*?



Bars: 95% intervals (adjusted row: cluster bootstrap over takers). Shaded: ±5 pp = no practical difference. Every check straddles zero.

BOUNDING THE NULL

Not significant isn't *no effect*

12.6 pp

SMALLEST DROP WE'D RELIABLY DETECT

80% power, one-tailed 5% test. The observed gap (−3.01 pp) sits inside this blind spot.

11.8 pp

BIGGEST DROP THE DATA CAN'T RULE OUT

Equivalence within ±5 pp fails (90% CI −11.8 to +4.0, p = 0.34), and so does ±10 pp (p = 0.07). ±5 pp is more than the established taker's +1.74 pp edge. Margins set after the analysis.

76.8%

CHANCE THE FOULED TAKER CONVERTS LESS

Bayesian, flat priors. 37.4% chance the drop exceeds 5 pp; 95% credible interval −13.7 to +5.2 pp.

The data lean towards a drop, can't rule out a large one, and can't rule out none.

Is it the foul, or the *taker*?

THE ESTABLISHED TAKER

+1.74 pp

Edge of the team's established taker (most penalties for the club before the match) over other takers: 95% CI −1.3 to +4.7. He took 35.4% of treatment kicks vs 30.6% of the rest.

THE TAKER'S RECORD

−0.08 reliability

Split-half, 126 takers with 10+ kicks (95% CI −0.47 to 0.18). A taker's hit rate on alternate kicks says nothing useful about the rest — penalty records behave like noise at these sample sizes.

Define the taker from the whole season instead and the edge jumps to +11.37 pp — largely the outcome leaking in: takers who miss lose the job.

2×2 CONTEXT

Where the *treatment cell* sits

	GOALKEEPER CAUSED THE FOUL	OUTFIELDER CAUSED THE FOUL
TAKER = THE PLAYER WHO WAS FOULED	TREATMENT 75.95% N = 79	79.02% N = 491
TAKER = NOT THE FOULED PLAYER (INCL. HANDBALLS)	78.33% N = 203	79.00% N = 2,762

Accent-bordered cell = treatment. The other three pool into control.

PER-LEAGUE REPLICATION

The same test, *league by league*

LEAGUE	T. RATE	T. N	C. RATE	C. N	Z	P (1-T)	P (HOLM)	SIG
Serie A	*80.00%*	*15*	78.57%	812	+0.134	0.5532	1.000	No · low n
La Liga	*55.56%*	*18*	77.37%	716	-2.167	0.0151	0.076	No · low n
Premier League	*91.30%*	*23*	79.63%	599	+1.375	0.9154	1.000	No · low n
Bundesliga	*60.00%*	*10*	79.83%	575	-1.539	0.0619	0.247	No · low n
Ligue 1	*84.62%*	*13*	79.71%	754	+0.437	0.6689	1.000	No · low n

Cells of 10–23. SIG = after a Holm correction for five tests (smallest adjusted p = 0.076).

TAKEAWAYS

Who should *care*

Coaches

TRAINING & TACTICS

EXPLORATORY

Being the team's established taker adds little: +1.7 pp over other takers (−1.3 to +4.7).

n = 1,085 vs 1,908 penalties · 95% CI includes zero · higher in 4 of 5 leagues

Managers

MATCH-DAY DECISIONS

EXPLORATORY

No case yet for taking the ball off a fouled player: −3.0 pp, plausibly anywhere from −13.6 to +5.2.

n = 79 vs 3,456 penalties · p = 0.26 · -3.25 pp after controls

Clubs

RECRUITMENT & CONTRACTS

MODERATE

Don't pay for a penalty record: a taker's hit rate on alternate kicks didn't predict the rest (reliability −0.08).

126 takers with 10+ kicks · split-half · 95% CI -0.47 to 0.18

Agents

POSITIONING A PLAYER

No defensible claim for this audience from this study.

EVIDENCE STRONG MODERATE EXPLORATORY Definitions: methodology § Evidence labels

LIMITATIONS

What this *can't* tell you

— Small treatment group: 79 penalties — a real drop below ~12.6 pp would usually go unseen.

— Taking it is a choice: status, record and experience are controlled; nerve and knocks on the day are not.

— "Fouled" is not "hurt": the data record who won the penalty, not whether he was injured.

— No minute, scoreline or goalkeeper in the model — the player-match table has neither.

— 575 of 4,110 penalties can't be linked (mostly a team taking 2+ in a match).

CONCLUSION

What the data *says*

The bench answer: *can't yet tell.*

Conversion is 3.01 pp lower when the fouled player takes the kick himself (75.95% vs 78.96% for everyone else). The point estimate moves in the predicted direction, but the treatment cell is too small to rule chance out (p = 0.258, one-tailed). Every check lands in the same place (−3.25 pp after controls). With 79 such penalties the design only reliably detects a drop of about 12.6 pp, and the data can't rule out one as large as 11.8 pp.

→ No statistical warrant to override the fouled player who wants to take it himself. Equally, no clean evidence he's at full strength.

Full methodology, source code and data:

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renball.com

/PROJECTS/01-PENALTY-ANALYSIS/

→ methodology formal H_0 / H_1, test assumptions, limitations

→ analysis.html interactive dashboard, live numbers from data.json

→ code/build.py Python pipeline — raw pickles → aggregated JSON